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SCHOLARLY REVIEW

Interaction Between Caffeine and Creatine When Used as Concurrent Ergogenic Supplements: A Systematic Review

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There is some controversy regarding the interactions between creatine (CRE) and caffeine (CAF) supplements. The aim of this systematic review was to study whether such ergogenic interaction occurs and to analyze the protocol to optimize their synchronous use. The PubMed, Web of Science, MEDLINE, CINAHL, and SPORTDiscus databases were searched until November 2021 following the PRISMA guidelines. Ten studies were included. Three studies observed that CRE loading before an acute dose of CAF before exercise did not interfere in the beneficial effect of CAF, whereas one study reported that only an acute supplementation (SUP) of CAF was beneficial but not the acute SUP of both. When chronic SUP with CRE + CAF was used, two studies reported that CAF interfered in the beneficial effect of CRE, whereas three studies did not report interaction between concurrent SUP, and one study reported synergy. Possible mechanisms of interaction are opposite effects on relaxation time and gastrointestinal distress derived from concurrent SUP. CRE loading does not seem to interfere in the acute effect of CAF. However, chronic SUP of CAF during CRE loading could interfere in the beneficial effect of CRE.

Keywords: ergogenic aids, coffee, athletic performance, sport nutrition

Caffeine (CAF) and creatine (CRE) are two supplements with high levels of effectiveness and safety (Kerksick et al., 2018; Maughan et al., 2018). Both supplements are widely used by athletes for both competitive and recreational use (Maughan et al., 2018; Pakulak et al., 2021), and given their popularity, there has been increasing interest in their concurrent use (Trexler & Smith-Ryan, 2015).

The CRE supplementation (SUP) has been shown to have a beneficial effect on repeated high-intensity exercise, muscle mass, muscle strength, recovery, and glycogen synthesis (Inácio et al., 2016; Kreider et al., 2017; Maughan et al., 2018). During and after intense exercise, the energy supplied to rephosphorylate adenosine diphosphate depends largely on the amount of phosphocreatine (PCr) stored in muscle, which becomes depleted during intense exercise (Schlattner et al., 2016; Ydfors et al., 2016). Therefore, CRE SUP can increase PCr storage in muscle, allowing an accelerated rate of adenosine triphosphate resynthesis during high-intensity, short-duration exercise (Greenhaff et al., 1993; Harris et al., 1992). The response to CRE SUP depends on the individual's initial PCr storage. The higher the PCr storage, the smaller the increase experienced with CRE intake. Thus, the effectiveness will be greater in less trained individuals or at the beginning of workload seasons.

In a normal diet that contains 1–2 g per day of CRE, muscle CRE stores are about 60%–80% saturated. Therefore, dietary CRE SUP serves to increase muscle CRE and PCr by 20%–40% (Kreider et al., 2017). The protocol of CRE SUP established by the International Olympic Committee consists of a loading phase with 20–25 g per day for 5–7 days followed by a maintenance phase of 3–5 g per day, whereas International Society of Sports Nutrition protocols have no loading phase and appear to be equally effective (Kreider et al., 2017).

The CAF SUP has been shown to improve performance in aerobic and anaerobic sports (Evans et al., 2018). Different mechanisms have been proposed to explain the ergogenic effect of CAF, but the main mechanism appears to be its effect as an adenosine receptor antagonist (Mielgo-Ayuso et al., 2019). By inhibiting adenosine action, CAF can enhance motivation and alertness while reducing fatigue perceptions and muscle pain (Ehlert et al., 2020).

Low to moderate doses, between 3 and 6 $mg \cdot kg^{-1}$, consumed 60 min before exercise are recommended by current guidelines (Goldstein et al., 2010; Guest et al., 2021; Maughan et al., 2018; Pickering & Kiely, 2018), whereas doses above 9 $mg \cdot kg^{-1}$ provide no benefit and may cause side effects such as gastrointestinal distress, nervousness, insomnia, or anxiety. However, there are secondary interindividual differences that may affect the efficacy of CAF SUP. These interindividual differences include habituation, training status, and method of CAF administration or genetic predispositions (Pickering & Kiely, 2018).

Due to the large evidence supporting the ergogenic effect of CAF and CRE, many nutritional supplements include both substances. Nevertheless, this combination is controversial as although they appear to have no pharmacokinetic interaction (Vanakoski et al., 1998) and increase performance through different mechanisms (Maccaferri et al., 2012), some studies have shown that CAF may decrease the effectiveness of CRE (Hespel et al., 2002; Quesada & Gillum, 2013; Vandenberghe et al., 1996).

Considering the existing controversy of the concomitant use of those substances, the objective of this systematic review was to analyze the effectiveness of concurrent CAF and CRE SUP on physical performance, to test whether the two supplements interact with each other, and if they do, to study what doses of both supplements may allow their simultaneous use.

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Methodology

This systematic review has been developed following the Preferred Reporting Items for Systematic Reviews and Meta-analysis recommendations: the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher et al., 2009).

Inclusion/Exclusion Criteria

The eligibility criteria considered for the studies' inclusion were the PICOS model: population, intervention, comparators, outcomes, and study design. The population studied was composed of healthy female and male subjects of any age and level of sport who were physically inactive, physically active, or professional athletes. The interventions considered were chronic or acute SUP with a defined dose of a combination of CRE and CAF and any form of acute or chronic standardized physical exercise after acute or chronic SUP period. As comparators, the effects between placebo (PLA) versus CRE and CAF were analyzed. Outcomes reported were body weight (BW), heart rate (HR), oxygen consumption (VO_2), maximal oxygen consumption (VO_2max), one repetition maximum (1RM), epinephrine (Ep), norepinephrine concentrations, serum free fatty acid, cardiopulmonary responses, respiratory exchange ratio, PCr, adenosine triphosphate, lactate, glucose, muscle power, muscle strength, rating of perceived exertion (RPE), time to exhaustion (TTE), ventilatory thresholds, peak power (PP), total work, repetitions to fatigue (RTF), relaxation time (RT), contraction time, electromyographic activity, and maximal torque. Finally, the design of the included studies should be partially, single, or double blind and randomized.

The following studies were excluded: animal studies, in vitro studies, studies conducted in injured or ill subjects, studies in which CRE and CAF were administered with other supplements, articles with no full-text available, articles not published in English or Spanish, opinion pieces, review articles, commentaries, and editorials.

Literature Search

The sources of information were obtained by searching for studies using the PubMed, Web of Science, MEDLINE Complete, CINAHL, and SPORTDiscus databases. The terms used for the literature search were: (("caffeine") AND ("creatine") AND ("ergogenic" OR "supplementation")). All original articles published until November 5th, 2021, were considered.

Study Selection

A two-stage search strategy was employed after duplicates were removed. In the initial stage, titles and abstracts were screened to exclude irrelevant articles according to eligibility criteria. At this stage, articles of questionable suitability were included. In the second stage, full texts of the investigations identified in the first stage were read to see whether they met the inclusion criteria. Reference sections of relevant articles were also examined via the snowball strategy.

Data Extraction

The following information was extracted from the selected articles: study source (authors and year of publication), participant characteristics (number of participants, sex, age, level of activity, and sports discipline), intervention protocol (type of exercise, SUP type and dose, and period), and outcomes (any factor to evaluate the improvement of physical performance).

Quality Assessment and Risk of Bias

The quality of each investigation was assessed following the Cochrane Collaboration Guidelines. The Cochrane Risk of Bias tool for randomized clinical trials is based on seven domains: sequence generation and allocation concealment (selection bias), blinding of participants and personnel (performance bias), binding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting (reporting bias), and other sources of bias (other bias). Risk of bias was categorized as low, high, or unclear.

Results

Study Selection

Through the database searches, 571 records were identified. Six additional studies were identified through other sources. Of these records, 311 duplicates were removed and 230 articles excluded after screening the title and abstract for eligibility. This left 36 studies, which were assessed for eligibility. Of these 36 studies, another 26 were removed according to the inclusion/exclusion criteria. Ten studies were finally included in this systematic review (Figure 1).

Characteristics of the Studies

The studies finally included were 10 randomized PLA-controlled trials, seven double blind (Doherty et al., 2002; Hespel et al., 2002; Lee et al., 2011, 2012; Pakulak et al., 2021; Vanakoski et al., 1998; Vandenberghe et al., 1996), two single blind (Jerônimo et al., 2017; Quesada & Gillum, 2013), and a partially blind trial (Trexler et al., 2016). Eight of the 10 studies included had a crossover design (Doherty et al., 2002; Hespel et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Quesada & Gillum, 2013; Vanakoski et al., 1998; Vandenberghe et al., 1996), whereas the remaining studies had a parallel design (Pakulak et al., 2021; Trexler et al., 2016). In total, there were 170 participants in the 10 studies (157 men and 13 women). Eight of the 10 studies included had fewer than 20 participants (Doherty et al., 2002; Hespel et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Quesada & Gillum, 2013; Vanakoski et al., 1998; Vandenberghe et al., 1996); only two assessed the effects of CAF and CRE in a larger group of subjects (Pakulak et al., 2021; Trexler et al., 2016). Most of the studies included participants aged between 18 and 25 years. However, in three studies, participants' age range was up to about 30 years (Doherty et al., 2002; Jerônimo et al., 2017; Vanakoski et al., 1998).

In one study, more than 8 days of CRE SUP was administered (Pakulak et al., 2021). However, in the rest of the studies, the SUP period was between 5 and 8 consecutive days before test (Doherty et al., 2002; Hespel et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Trexler et al., 2016; Vanakoski et al., 1998; Vandenberghe et al., 1996) except for one study wherein the SUP protocol was acute (Quesada & Gillum, 2013). Daily CAF SUP given was 5–7 $mg \cdot kg^{-1}$ per day in eight studies (Doherty et al., 2002; Hespel et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Quesada & Gillum, 2013; Vanakoski et al., 1998; Vandenberghe et al., 1996), whereas the remaining studies used a lower dosage (Pakulak et al., 2021; Trexler et al., 2016). In two of

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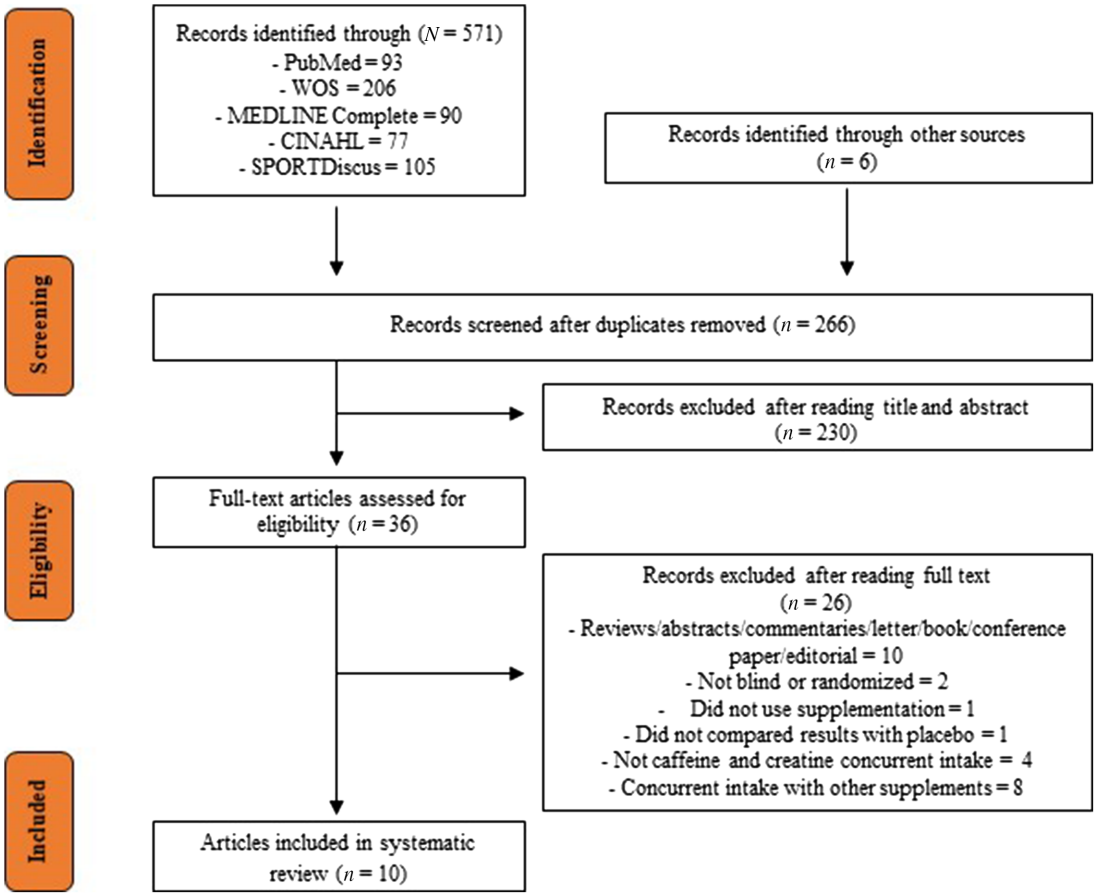

Figure 1 — Flow diagram of literature search according to the PRISMA statement.

the studies, the daily CRE SUP was 0.3 g·kg^{-1} four times a day for 5 days (Lee et al., 2011, 2012). The rest of the studies used different protocols for CRE SUP: 0.1 g·kg^{-1} per day for 6 weeks (Pakulak et al., 2021), 0.3 g·kg^{-1} four times a day for 6 days (Doherty et al., 2002), 0.1 g·kg^{-1} three times a day for 3 days (Vanakoski et al., 1998), 3 g for 7 days (Jerônimo et al., 2017); four intakes of 20 g for 5 days (Trexler et al., 2016), four intakes of 5 g for 4 days (Hespel et al., 2002), 0.5 g·kg^{-1} for 6 days (Vandenberghe et al., 1996), and acute intake of 100 mg·kg^{-1} 2 hr before the running protocol (Quesada & Gillum, 2013).

Quality Assessment and Risk of Bias

In all studies, randomized sequence generation was categorized as low risk bias, and allocation concealment was unclear risk in five studies (Doherty et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Trexler et al., 2016) and low risk in the other five studies (Hespel et al., 2002; Pakulak et al., 2021; Quesada & Gillum, 2013; Vanakoski et al., 1998; Vandenberghe et al., 1996). In terms of performance bias, seven studies were characterized as low risk (Doherty et al., 2002; Hespel et al., 2002; Lee et al., 2011, 2012; Pakulak et al., 2021; Vanakoski et al., 1998; Vandenberghe et al., 1996), two as unclear (Jerônimo et al., 2017; Trexler et al., 2016), and the last one as high risk of bias (Quesada & Gillum, 2013). In detection bias, seven studies were characterized as low risk (Doherty et al., 2002; Hespel et al., 2002; Lee et al., 2011, 2012; Pakulak et al., 2021; Vanakoski et al., 1998; Vandenberghe et al., 1996) and three studies as high risk (Jerônimo et al., 2017; Quesada & Gillum, 2013; Trexler et al., 2016).

For attrition and reporting bias, six articles were categorized as low risk (Doherty et al., 2002; Pakulak et al., 2021; Quesada & Gillum, 2013; Trexler et al., 2016; Vanakoski et al., 1998; Vandenberghe et al., 1996) and four as unclear risk (Jerônimo et al., 2017; Lee et al., 2011, 2012; Vandenberghe et al., 1996). Finally, two studies were described as having an overall high risk of bias (Jerônimo et al., 2017; Quesada & Gillum, 2013) and eight a low risk of bias (Doherty et al., 2002; Hespel et al., 2002; Lee et al., 2011, 2012; Pakulak et al., 2021; Trexler et al., 2016; Vanakoski et al., 1998; Vandenberghe et al., 1996). This information is detailed in Table 1 and Figure 2.

Results of Individual Studies

To analyze this controversial issue, the 10 studies included can be classified into three groups: (a) studies that used an acute SUP of both CAF and CRE (Quesada & Gillum, 2013), (b) studies that used an acute intake of CAF after a CRE loading (Doherty et al., 2002; Lee et al., 2011, 2012), and (c) finally, articles wherein CAF and CRE were supplemented chronically (Hespel et al., 2002; Jerônimo et al., 2017; Pakulak et al., 2021; Trexler et al., 2016;

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Table 1 Risk of Bias Graph

Study	Sequences generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcomes assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective outcome reporting (reporting bias)	Other sources of bias
Jerônimo et al. (2017)	+	?	?	−	?	?	−
Trexler et al. (2016)	+	?	?	−	+	+	+
Quesada and Gillum (2013)	+	+	−	−	+	+	−
Lee et al. (2012)	+	?	+	+	?	?	+
Lee et al. (2011)	+	?	+	+	?	?	+
Pakulak et al. (2021)	+	+	+	+	+	+	+
Doherty et al. (2002)	+	?	+	+	+	+	+
Hespel et al. (2002)	+	+	+	+	?	?	+
Vandenberghe et al. (1996)	+	+	+	+	+	+	+
Vanakoski et al. (1998)	+	+	+	+	+	+	+

Note. + = low bias risk; ? = unclear bias risk; − = high bias risk.

Vanakoski et al., 1998; Vandenberghe et al., 1996). Table 2 summarizes all the studies reviewed examining the concurrent CAF and CRE SUP on physical performance.

Acute SUP of CRE and CAF

Quesada and Gillum (2013) performed a single-blind randomized controlled trial (RCT) with a crossover design with seven moderately active men (20.8 ± 1.7 years). Two hours before an endurance exercise protocol, which consisted of running at 6 miles·hr^{-1} and increasing 0.6 miles·hr^{-1} every 3 min until volitional exhaustion, subjects took the CRE SUP (100 mg·kg^{-1}), and 20 min before the test they took the CAF SUP (6 mg·kg^{-1}). The trials were scheduled with a minimum interval of 72 hr with no more than 7 days between trials. The results showed a higher anaerobic ventilatory threshold ($p < .05$) after CAF intake compared with PLA, but there were no differences between CRE + CAF and PLA or CAF and CRE + CAF. There were no significant differences in VO_2max, RPE, perceived stress index, HR, and TTE among the three groups. In addition, three participants reported mild gastrointestinal discomfort during CRE + CAF trial.

Acute Intake of CAF After CRE Loading

Doherty et al. (2002) conducted a double-blind RCT with a crossover study design with 14 trained men (22.7 ± 3.5 years). Subjects performed treadmill running to volitional exhaustion at an exercise intensity equivalent to 125% of VO_2max. This study had three trials; the first one was before a period of 6 days of CRE loading (0.3 g·kg^{-1} per day of CRE; baseline [BASE]), and the subsequent two trials were after CRE SUP (12–24 hr after; 4–6 days after). One hour before the exercise protocol, the subjects ingested CAF (5 mg·kg^{-1}) or PLA. Body mass increased after CRE SUP ($p < .05$) without differences between CRE + CAF and CRE groups. Total VO_2 was significantly increased in the CRE + CAF group compared with CRE group ($p < .05$); nonetheless, no differences with BASE were reported. Moreover, the CRE + CAF group showed a tendency to decrease RPE at all time points. However, it was only significant at 90 s ($p < .05$). Regarding TTE, it was increased ($p < .05$) in CRE + CAF group when compared with BASE and CRE groups. Maximal accumulated oxygen deficit was nonsignificantly increased in the CRE + CAF group compared with the BASE and CRE groups. Finally, epinephrine and plasma glucose were higher in the CRE + CAF group compared with BASE and CRE groups ($p < .05$), and no differences between groups were found in lactate or in plasma potassium.

Both Lee et al. (2011) and Lee et al. (2012) completed a double-blind RCT with a crossover study design with 12 physically active men (19 ± 0.6 years and 20 ± 1.8 years, respectively). A similar SUP protocol consisting of either SUP of a CRE loading for 5 days (0.3 g·kg^{-1} per day) followed by an ingestion of CAF (6 mg·kg^{-1}) or PLA 1 hr before testing. In both studies, a BASE trial without any ergogenic aids was performed. The trial conducted by Lee et al. (2011) consisted of six high-intensity intermittent sprints of 10 s each on a cycle ergometer with 60 s of rest between sprints. They observed no differences between groups for BW, RPE, Ep, and norepinephrine. Nevertheless, authors found that CRE + CAF group increased mean and PP compared with CRE ($p < .05$) and with BASE ($p < .05$). Also, the mean and PP in the CRE group during a sprint was significantly greater in comparison with BASE ($p < .05$). In addition, plasma glucose and lactate increased with CRE + CAF SUP during most sprints and the posttest compared with CRE and BASE ($p < .05$). Moreover, HR increased during some sprints in CRE + CAF compared with BASE ($p < .05$) and CRE ($p < .05$). In

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Table 2 Summary of the Studies Reviewed Examining the Concurrent SUP With CAF and CRE on Physical Performance

Study	Design	Subjects	Supplementation groups	Exercise protocols	Variables	Results
Quesada and Gillum (2013)	Single-blind RCT with cross-over design	7 moderately active men (20.8 ± 1.7 years)	Three groups: PLA, CRE + CAF, and CAF + PLA Each trial was composed of: - First drink 2 hr before exercise: 100 mg·kg^{-1} of CRE or PLA - Second drink 20 min before exercise: 6 mg·kg^{-1} of CAF or PLA	3-min intervals, which increased by 0.6 miles·hr^{-1}, starting with 6 miles·hr^{-1} and ending at volitional exhaustion	VO_2max, anaerobic VT, HR, RPE, and TTE	VO_2max, HR, RPE, and TTE: CRE + CAF = CAF + PLA = PLA Anaerobic VT: ↑ in CAF + PLA compared with PLA CRE + CAF = CAF + PLA = PLA (There was interaction between substances)
Doherty et al. (2002)	Double-blind RCT with cross-over design	14 trained men (22.7 ± 3.5 years)	Three groups: BASE, CRE + CAF, and CRE + PLA CRE: 0.3 g·kg^{-1} per day for 6 days CAF or PLA: 5 mg·kg^{-1} 1-hr pretest	3 × 6 min run increase (80%, 85%, and 90% VO_2max) with 10.5% inclination and 5-min rest between runs	Body mass, VO_2, MAOD, RPE, TTE, Ep, glucose, potassium, and lactate	Body mass: ↑ in all groups BASE = CRE + CAF = CRE + PLA MAOD, potassium, and lactate: BASE = CRE + CAF = CRE + PLA VO_2: CRE + CAF > CRE + PLA RPE: BASE = CRE + PLA > CRE + CAF at 90 s after exercise TTE, Ep, and glucose: ↑ in CRE + CAF in comparison with BASE and CRE + PLA (There was no interaction between substances)
Lee et al. (2011)	Double-blind RCT with cross-over design	12 physically active men (19 ± 0.6 years)	Three groups: BASE, CRE + CAF, and CRE + PLA CRE: 0.3 g·kg^{-1} per day for 5 days continued with CAF or PLA: 6 mg·kg^{-1}	6 × 10 s IHIS	BW, glucose, HR, MP, PP, plasma lactate, RPE, Ep, and Nep	BW, RPE, Ep, and Nep: BASE = CRE + CAF = CRE + PLA Lactate and glucose: CRE + CAF > CRE + PLA = BASE MP and PP: CRE + CAF > CRE + PLA > BASE (There was no interaction between substances)
Lee et al. (2012)	Double-blind RCT with cross-over design	12 physically active men (20 ± 1.8 years)	Three groups: BASE, CRE + CAF, and CRE + PLA CRE: 0.3 g·kg^{-1} per day for 5 days CAF or PLA: 6 mg·kg^{-1}1-hr pretest	Pedal at a work rate of 50 W on a cycling ergometer. Every 2 min, increase by 30 W until maximal effort.	TTE, BW, HR, RPE, Ep, Nep, FFA, lactate, glucose cardiopulmonary responses, RER, and VO_2max	RER, VO_2max, Ep, Nep, and FFA: BASE = CRE + CAF = CRE + PLA RPE: CRE + CAF < CRE + PLA = BASE HR, glucose, lactate, and TTE: CRE + CAF > CRE + PLA = BASE (There was no interaction between substances)

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Table 2 *(continued)*

Study	Design	Subjects	Supplementation groups	Exercise protocols	Variables	Results
Pakulak et al. (2021)	Double-blind RCT with parallel design	28 physically active men (18) and women (10) (18–38 years)	Four groups: PLA, CAF, CRE, and CRE + CAF PLA: PLA CAF: 3 mg·kg^{-1} per day for 6 weeks CRE: 0.1 g·kg^{-1} per day for 6 weeks CRE + CAF: 0.1 g·kg^{-1} per day for 6 weeks of CRE + 3 mg·kg^{-1} per day for 6 weeks of CAF	Exercise days (3 sets to volitional fatigue per exercise) Day 1: chest and biceps muscles Day 2: leg and core muscles Day 3: rest day Day 4: back and triceps muscles Day 5: shoulder and core muscles Day 6: rest day This cycle was repeated for 6 weeks	RPE, BW, leg and chest press 1RM, leg and chest press endurance	RPE, leg and chest press 1RM, and leg and chest press endurance: PLA = CRE = CAF = CRE + CAF BW: - Females: Pre > post for CRE and CAF Pre = post for PLA and CRE + CAF - Males: Pre < post for CRE Pre = post for PLA, CAF, and CRE + CAF (There was no interaction between substances)
Vanakoski et al. (1998)	Double-blind RCT with cross-over design	8 trained men (6) and women (2) (18–29 years)	Four groups: PLA, CAF, CRE, and CRE + CAF PLA: PLA CAF: 7 mg·kg^{-1} for 1 day CRE: 300 mg·kg^{-1} per day for 3 days CRE + CAF: 300 mg·kg^{-1} per day for 3 days of CRE + 7 mg·kg^{-1} of CAF	2 protocols by an electronic ergometer: Anaerobic: 3 × 1 min; rest: 5 min Aerobic: 45-min constant workload (70 rpm)	TW, HR, glucose, lactate, and MPS	MPS, TW, HR, lactate, and glucose: PLA = CAF = CRE = CRE + CAF (There was no interaction between substances)
Vandenberghe et al. (1996)	Double-blind RCT with cross-over design	9 physically active men (20–23 years)	Three groups: PLA, CRE, and CRE + CAF 8 days of SUP CRE: 0.5 g·kg^{-1} per day for 8 days (6 days) PLA: PLA CRE + CAF: 0.5 g·kg^{-1} per day CRE (6 days) + (3 days) 5 mg·kg^{-1} per day CAF	2-min rest intervals of: 3 static MVCs of the knee extension 3 bouts of 30 dynamic MVCs 4 bouts of 20 dynamic MVCs 5 bouts of 10 dynamic MVCs (Bouts of 30, 20, and 10 MVCs were separated by 60, 40, and 20 s, respectively)	BW, PCr concentration, static torque, and dynamic torque	BW and static torque: PLA = CRE = CRE + CAF PCr: CRE + CAF = CRE > PLA Dynamic torque: ↑ Dynamic Torque in CRE ↔ Dynamic Torque in CRE + CAF (There was interaction between groups)

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Table 2 *(continued)*

Study	Design	Subjects	Supplementation groups	Exercise protocols	Variables	Results
Hespel et al. (2002)	Double-blind RCT with cross-over design	10 physically active men (9) and women (1) (21–24 years)	Five groups: PLA, CRE, acute CAF, chronic CAF, and CRE + CAF PLA: PLA (7 days) CRE: 4 days pretest 4 × 5 g per day Acute CAF: 1 hr pretest 5 mg·kg^{-1} Chronic CAF: 3 days pretest 5 mg·kg^{-1} CRE + CAF: 3 days pretest 4 × 5 g per day CRE + 5 mg·kg^{-1} CAF	3 × 3 s (90°) voluntary maximal isometric knee extensions (10-s rest intervals)	RT, CT, and Tmax	CT and Tmax: No differences between groups RT: CRE < PLA = acute CAF = CRE + CAF < chronic CAF (There was interaction between substances)
Trexler et al. (2016)	Partially blind RCT with parallel design	54 physically active men (21.1 ± 2.1 years)	Four groups: PLA, CRE, CRE + PLA, and CRE + COF 5 days of SUP Pretest: CRE: 20 g per day CRE + CAF: 20 g per day CRE + 300 mg per day CAF CRE + COF: 20 g per day CRE + 8.9 g per day COF PLA: 3.8 g per day	2 protocols Strength test: Adjust load to 1RM (2- to 3-min rest between attempts) 3-min rest, RTF test (lift the 80% 1RM maximal repetition) (10 min after) Repeated sprint protocol: 5 × 10 s sprint on a cycle ergometer 95 g·kg^{-1} + weight basket (60-s rest between sprints)	Serum CRE levels, muscle strength, 1RM, RTF, PP, TW, and BW	1RM, PP, TW, and BW: PLA = CRE = CRE + CAF = CRE + COF RTF: - Leg press: pre < post in all groups - Bench press: pre < post in all groups except for CRE + COF group (There was no interaction between substances)
Jerônimo et al. (2017)	Single-blind RCT with cross-over design	16 physically active men (18–30 years)	Four groups: PLA, CRE, CAF, and CRE + CAF CAF: 6 mg·kg^{-1} for 3 days + detox 5 days CRE: + 3 g for 7 days CRE + CAF: 3 g CRE + 6 mg·kg^{-1} CAF for 3 days	45 repetitions knee extension and flexion (120°·s^{-1} angular speed) on Biodex	EMG activity, torque, and fatigue	EMG: ↑ CAF (4.57%) and CRE + CAF (3.07%) ↓ CRE (17.07%) Torque: ↑ CAF (4.25%), CRE (3.45%), and CRE + CAF (5.79%) (There was synergy between substances)

Note. BASE = baseline; BW = body weight; CAF = caffeine; CAF + PLA = caffeine SUP plus placebo; COF = coffee; CT = contraction time; Con = control; CRE = creatine; CRE + CAF = creatine plus caffeine SUP; CRE + PLA = creatine SUP plus placebo; EMG = electromyography; Ep = epinephrine; FFA = serum free fatty acid; HR = heart rate; IHIS = intermittent high-intensity sprint; MAOD = maximal accumulated oxygen deficit; MP = mean power; MPS = maximum pedal speed; MVCs = maximal voluntary contractions; Nep = norepinephrine; PP = peak power; PCr = phosphocreatine; PLA = placebo; RM = repetition maximum; RPE = rating of perceived exertion; rpm = revolutions per minute; RCT = randomized controlled trial; RT = relaxation time; RTF = repetitions to fatigue; RER = respiratory exchange ratio; SUP = supplementation; Tmax = maximal torque; TTE = time to exhaustion; TW = total work; VO_2max = maximal oxygen consumption; VO_2 = oxygen consumption; VT = ventilatory threshold.

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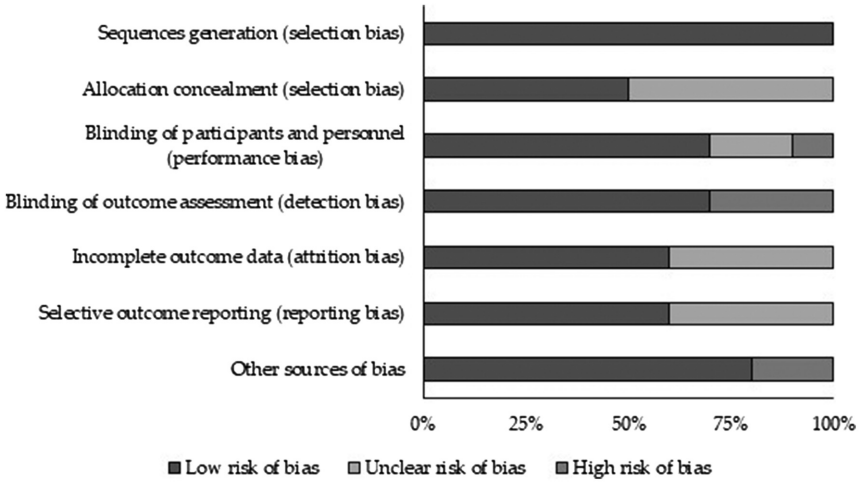

Figure 2 — Risk of bias summary.

the exercise protocol by Lee et al. (2012), the subjects began pedaling at a rate of 50 W on a cycloergometer, increasing 30 W every 2 min. They reported that intake of CRE + CAF increased TTE, maximal HR, blood glucose, and blood lactate and decreased RPE at 230 W compared with BASE and CRE groups. The free fatty acids were increased in CRE + CAF at postingestion compared with CRE but not at pretest or posttest. No differences were observed between groups for BW, respiratory exchange ratio, relative and absolute VO_2max, Ep, and norepinephrine.

Chronic Intake of CAF and CRE

Pakulak et al. (2021) completed a double-blind RCT with a parallel design with 28 healthy active participants (18 men and 10 women). Subjects were randomly divided into four groups for a 6-week period of SUP: PLA (four men and two women; 23 ± 7 years), CRE (five men and two women; 22 ± 4 years; 0.1 $g{\cdot}kg^{-1}$ per day), CAF (three men and three women; 19 ± 1 years; 3 $mg{\cdot}kg^{-1}$ per day), and CRE + CAF (six men and three women; 22 ± 4 years; 0.1 $g{\cdot}kg^{-1}$ per day + 3 $mg{\cdot}kg^{-1}$ per day, respectively). Participants followed the same resistance training program for the 6 weeks of SUP. The training program started the first day of SUP and was divided into a 6-day cycle, which was repeated each week for the entire intervention. From Days 1–5, subjects performed a resistance training that consisted of three sets to volitional fatigue per exercise (exercises differed between days) with 2-min rest between sets. Days 3 and 6 served as rest days from training. There were no significant differences between the four groups for average RPE or average fatigue per training session. In addition, no significant differences between groups were observed for muscle strength, measured as leg and chest press 1RM, and muscle endurance, measured as total repetitions performed for leg and chest press endurance (repetitions until volitional fatigue using 50% BASE 1RM). Nevertheless, there was a Group × Sex × Time interaction for BW ($p = .034$). Women in CRE and CAF groups experienced a decrease in BW from pre- to postintervention, whereas men in CRE group had an increase in BW from pre- to postintervention.

Vandenberghe et al. (1996) carried out a double-blind RCT with a crossover study design in nine physically active men (20–23 years) who were assigned in random order to three experimental protocols with a washout period of 3 weeks between them. In each protocol, the following SUP was given for 6 days: CRE (0.5 $g{\cdot}kg^{-1}$ per day of divided into eight doses), PLA or CRE (0.5 $g{\cdot}kg^{-1}$ per day of divided into eight doses + 5 $mg{\cdot}kg^{-1}$ per day) + CAF during the last 3 days (5 $mg{\cdot}kg^{-1}$ per day). Individuals performed three consecutive maximum isometric contractions and three interval series of 90, 80, and 50 maximum voluntary contractions performed with a 2-min rest interval between series. No significant changes were found in BW over the three groups. Dynamic torque production with CRE intake increased by 10%–23% ($p < .05$), whereas it did not change in CRE + CAF group. In the groups that ingested CRE, the muscle PCr concentrations increased by 4%–6% when compared with PLA group ($p < .05$); however, no differences were found between CRE and CRE + CAF groups.

Vanakoski et al. (1998) conducted a crossover RCT and double-blind study wherein eight subjects (two women and six men aged 18–29 years) were randomly assigned into four groups, receiving all the following treatments in random order: PLA, CAF (7 $mg{\cdot}kg^{-1}$), CRE (300 $mg{\cdot}kg^{-1}$ per day), or CRE + CAF (300 $mg{\cdot}kg^{-1}$ per day + 7 $mg{\cdot}kg^{-1}$ per day, respectively) for 3 days. In a first session without exercise, subjects ingested CAF before starting the SUP period. CRE or the corresponding PLA SUP was administered during 3 days prior to the test. The last dose of both CRE and CAF was administered 1 hr and 10 min before the trial. Subjects performed three repetitions of 1 min on a cycloergometer at maximum speed. After anaerobic exercise, individuals performed 45 min of pedaling at a constant aerobic speed. There was a 1-week separation between the sessions. Neither CRE, CAF, nor CRE + CAF improved the maximum pedal speed (rpm) or total work output during the 1-min series compared with PLA. In addition, no significant differences in HR, blood lactate, glucose, or hypoxanthine levels were found between groups for any of the exercise tests.

Hespel et al. (2002) performed a double-blind RCT with a crossover study design in 10 physically active subjects (nine men and one woman aged 21–24 years) who were divided into five experimental protocols. The subjects performed the exercise test before and after intaking CRE (20 g per day for 4 days), short-term intake of CAF (5 $mg{\cdot}kg^{-1}$ per day for 3 days), intake of CRE + CAF (20 g per day for 3 days of CRE + 5 $mg{\cdot}kg^{-1}$ per day of CAF for 3 days), acute intake of CAF (5 $mg{\cdot}kg^{-1}$ per day 1 hr before the

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test), or PLA. CRE group had lower RT when compared with PLA group ($p < .05$), whereas acute CAF group tended to be higher ($p = .09$). In contrast, chronic CAF group had higher RT than PLA, acute CAF, and CRE + CAF groups. No differences were observed for RT between PLA, acute CAF, and CRE + CAF groups. Finally, maximal torque and contraction time showed no differences between all groups.

Trexler et al. (2016) carried out a partially blind RCT with a parallel study design in 54 physically active men (20.1 ± 2.1 years). The test consisted of 1RM and RTF with 80% of 1RM for the bench press and leg press, followed by a five-spring repeat test of 10 s with 60 s of rest between sets on a cycloergometer. The test was performed before and after the SUP period. Seventy-two hours after the test, the subjects were randomly assigned into four groups: CRE (20 g per day), CRE + CAF (20 g per day of + 300 mg per day, respectively), CRE + coffee (4 × 5 g per day + 8.9 g, respectively), or PLA. The SUP was taken for 5 days, and after it, the physical test was performed. The results showed that 1RM increased in all groups for leg press and bench press ($p < .05$), and RTF increased in all groups for leg press, whereas for bench press, the CRE + coffee group did not increase. No differences between groups were found in RTF and PP ($p < .05$). When CRE SUP was ingested, serum creatinine levels increased significantly. Finally, four participants of the CRE + CAF group suffered gastrointestinal problems.

Jerônimo et al. (2017) completed a single-blind RCT with a crossover study design in 16 physically active men (18–30 years). After BASE test, which consisted of 45 repetitions of knee extension and flexion (120 s^{-1}) on a dynamometer, the subjects took 6 $mg \cdot kg^{-1}$ of CAF per day for 3 days, followed by 5 days of washout. After the washout period, subjects began 7 days of 3 g per day of CRE and finished the protocol by taking 3 g of CRE + 6 $mg \cdot kg^{-1}$ of CAF per day for 3 days. The results reported that CAF and CRE + CAF groups had higher electromyographic activity (4.57% and 3.07%, respectively), whereas in CRE group, it decreased by 17.07%. The torque increased significantly in all groups: 4.25% in the CAF group, 3.45% in the CRE group, and 5.79% in the CRE + CAF group. Fatigue decreased, although not significantly, in the CRE + CAF group compared with BASE. Also, no differences between groups were found in peak torque.

Discussion

This systematic review analyzes the effect of the coingestion of CAF and CRE SUP on sports performance in athletes. The three studies that examined the effect of acute CAF SUP after a CRE loading concluded that the ingestion of CAF maintained its ergogenic effect despite the previous intake of CRE loading (Doherty et al., 2002; Lee et al., 2011, 2012). However, when chronic SUP was administered, the results were controversial: one study determined synergy between CAF and CRE (Jerônimo et al., 2017), two studies observed interaction (Hespel et al., 2002; Vandenberghe et al., 1996), and three studies did not report interaction (Pakulak et al., 2021; Trexler et al., 2016; Vanakoski et al., 1998).

Both supplements enhance performance through different physiological mechanisms, and therefore, no pharmacokinetic interaction should occur when taken together. Nevertheless, previous research has suggested that there is an interaction between CAF and CRE (Harris et al., 2005). Several studies have found that subjects report gastrointestinal distress (Quesada & Gillum, 2013; Trexler et al., 2016), and some studies suggest that the relationship between CRE + CAF and exercise performance is conditioned by gastrointestinal problems. In particular, it appears that CAF attenuates the ergogenic effects of CRE SUP, although there is insufficient evidence to confirm this (Harris et al., 2005; Hespel et al., 2002; Vandenberghe et al., 1996).

When acute CAF was ingested after a CRE loading, total VO_2 and TTE increased, whereas RPE in a treadmill run to exhaustion decreased significantly compared with other groups (Doherty et al., 2002; Lee et al., 2012). The same results were obtained by Guest et al. (2021), who affirmed that pain attenuation during exercise as a result of CAF SUP could result in RPE decrease and, consequently, an increase in TTE. In terms of muscle strength, Jerônimo et al. (2017) observed that the torque production increased in the three SUP groups, but CRE + CAF group achieved significantly higher torque values. However, Vandenberghe et al. (1996) and Hespel et al. (2002) showed that CAF counteracted the ergogenic effect of CRE. Both studies showed that dynamic torque production was increased by CRE but was not changed by CRE + CAF. It is important to mention that in the study of Vandenberghe et al. (1996), the last dose of CAF they took preceded the exercise test by at least 20 hr, and given that the plasma half-life of CAF is 3–5 hr, it is reasonable that at the time of the exercise test, the substance would have been eliminated. Therefore, the effect of CAF counteracting the ergogenic action of CRE is unlikely to be explained by its effect on muscle energetics. Similar effects as those found by Vandenberghe et al. (1996) and Hespel et al. (2002) were observed by Trexler et al. (2016) and Pakulak et al. (2021), who observed no effects on strength or sprint performance between the SUP group against PLA. However, Pakulak et al. (2021) used exercises protocols that may not be limited by the phosphate energy system, so CRE SUP is not necessary in this situation. These results confirm that CRE SUP does not influence in the effects of CAF SUP on performance, whereas CAF SUP may blunt the effect of CRE SUP. This recurrent problem seems to be the turning point in the synchronous use of both supplements, and therefore, an adaptation period for the intake of both supplements is necessary to allow athletes to benefit from their intake simultaneously.

The RT was shortened after CRE SUP during intermittent maximal isometric muscle contraction (Van Leemputte et al., 1999) and after voluntary maximal isometric knee extensions (Hespel et al., 2002). However, Hespel et al. (2002) observed that CAF and CRE had opposite effects on muscle RT. Whereas CAF increased RT, CRE decreased it. These contrary effects can be explained through calcium management in the sarcoplasmic reticulum. Whereas CRE appears to facilitate sarcoplasmic reticulum Ca^{2+} reuptake, CAF inhibits sarcoplasmic reticulum Ca^{2+} adenosine triphosphatase, increasing calcium release (Hespel et al., 2002). This theory could also provide a mechanism to explain how CAF counteracted the ergogenic effect of CRE on RT, as found by Vandenberghe et al. (1996). Nevertheless, in this case, neither CRE availability nor muscle energy was affected (Vandenberghe et al., 1996; Figure 3).

The main limitation of this systematic review was the low sample size of the studies included. Eight studies had <20 participants (Doherty et al., 2002; Hespel et al., 2002; Jerônimo et al., 2017; Lee et al., 2011, 2012; Quesada & Gillum, 2013; Vanakoski et al., 1998; Vandenberghe et al., 1996), so statistical power was limited. In addition, another limitation was the low number of articles that assessed with randomized and blinded designs the effect of concurrent CRE and CAF SUP, so only 10 articles were included in this systematic review. A further limitation is the high heterogeneity in the methodology used in the SUP protocols and in the types of exercise they have used to measure the variables in the different studies. Moreover, there have not been many studies

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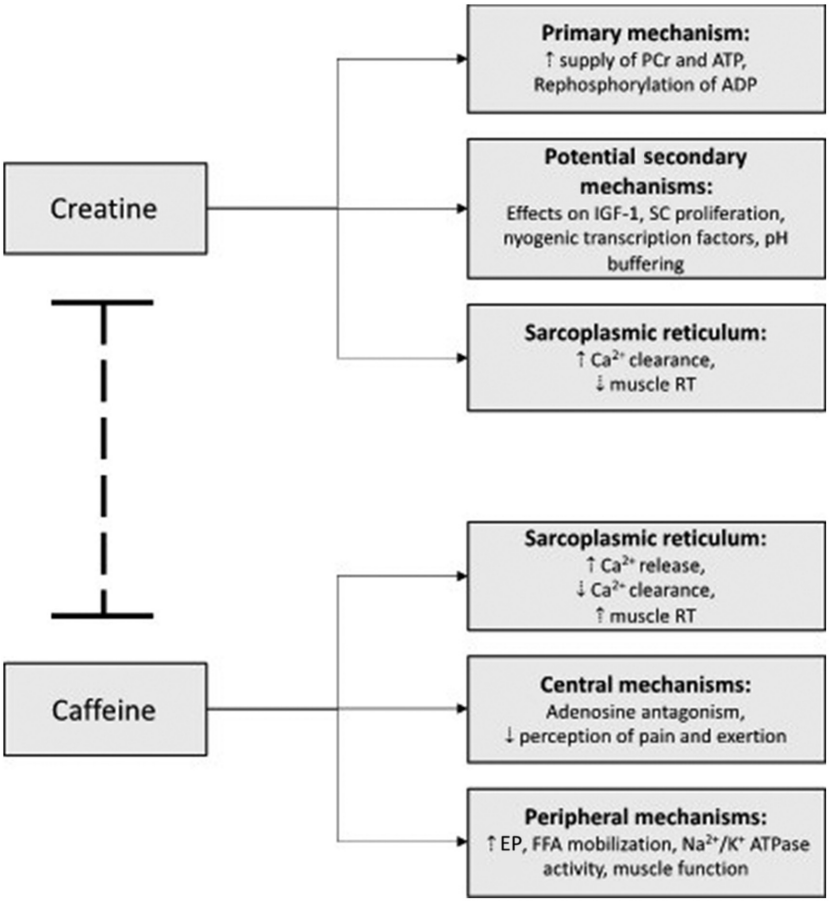

Figure 3 — Mechanisms and potential sources of interference of caffeine and creatine. PCr = phosphocreatine; ATP = adenosine triphosphate; ADP = adenosine diphosphate; IGF-1 = insulin-like growth factor 1; SC = satellite cells; RT = muscle relaxation time; Ep = epinephrine; FFA = free fatty acids. Adapted from "Creatine and caffeine: Considerations for concurrent supplementation," by Trexler, E.T. & Smith-Ryan, A.E. 2015, *International Journal of Sport Nutrition and Exercise Metabolism, 25*(6), pp. 607–623.

analyzing the concurrent effect of the CRE and the CAF in the absence of any other supplement.

To provide an appropriate guideline for the simultaneous use of both supplements, and after having conducted this systematic review, we can conclude that (a) the combination of acute doses of CRE with acute doses of CAF showed no effect on exercise performance except for anaerobic ventilatory thresholds; (b) CRE loading when combined with acute CAF SUP prior to exercise does not impair performance and in some cases improves it (RPE, TTE, power, and glucose); and (c) CRE loading combined with chronic doses of CAF SUP could interfere with the ergogenic effect of both substances and, thereby, affect certain performance-related variables (RPE, RT, dynamic torque, and electromyographic activity). In addition, it should be noted that the synchronous use of CRE and CAF can produce gastrointestinal distress, leading to decreased physical performance.

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